

Muhammad Achad Ansori

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Electronics Engineer specializing in embedded systems, PCB design, and microcontroller development. Experienced across the full product development cycle, from schematic and PCB layout to firmware programming, EMC troubleshooting, and testing and validation. Strong foundation in STM32/ESP32 programming, microcontroller peripheral interfacing, and version-controlled embedded workflows.

Education

Bachelor of Applied Science, Electronics Engineering	Aug 2021 - Aug 2025
Politeknik Elektronika Negeri Surabaya (PENS)	Surabaya, Indonesia
- Advisor: Arif Irwansyah - Final Project: "Jaladhi, Autonomous Surface Vehicle with Collision Avoidance". Embedded system integration on Jetson Nano with ROS, computer vision, and multi-sensor fusion (camera and RPLIDAR). Awarded 2nd Place, PENS Final Project Competition	
Machine Learning Cohort, Bangkit Academy	Google, Tokopedia, Gojek, Traveloka
Aug 2024 - Dec 2024	
Completed 5-month intensive program in Machine Learning, Deep Learning, TensorFlow, and Computer Vision	

Professional Appointments

Research and Development Engineer · Contract	PT Aldzama	Feb 2026 - Present
<i>Promoted from Demolition Robot Technician based on demonstrated engineering capability.</i>		
- Designed custom embedded control unit for demolition robots from scratch using STM32, reducing dependency on expensive OEM replacement parts - Developed firmware in C/C++ utilizing microcontroller peripherals including UART, CAN bus, PWM, and digital I/O for real-time motor and actuator control - Reverse-engineered and integrated the original remote control system with the in-house controller - Designing electro-hydraulic control modules, interfacing embedded control signals with proportional valve and actuator systems - Performed testing, validation, and field troubleshooting of electronic and control systems at client sites		
Demolition Robot Technician · Contract	PT Aldzama	Sep 2025 - Jan 2026
- Developed a custom control unit for demolition robots from scratch, significantly reducing dependency on expensive OEM replacement parts - Reverse-engineered and modified the original remote control system to integrate with the in-house built controller - Promoted to R&D Engineer role after 3 months based on demonstrated engineering capability		
Electronics Engineer · Internship	PT Venambak Kail Dipantara	Dec 2024 - May 2025
- Redesigned PCB layout with ESP32 for FeederVE control box, handling the full production cycle from schematic and PCB design to assembly and panel box integration for 50+ units - Resolved electromagnetic interference (EMC/EMI) issues on the BoxVE system through PCB redesign, ensuring stable operation in outdoor environments - Developed embedded software for AutoVE (autonomous mobile aerator) using Python on a low-cost mini PC		
Maintenance and Improvement Engineer · Internship	PT Toshin Prima Fine Blanking (Indoprima Group)	Feb 2024 - Jul 2024
- Developed and deployed YOLOv4-based visual inspection system on Jetson Nano achieving ~29 FPS for real-time OK/NG quality control detection - Built IoT monitoring system for 2 critical factory chiller units using Raspberry Pi with web dashboard, enabling real-time monitoring to prevent factory-wide downtime - Performed routine maintenance and repair of industrial fine blanking machines		
ROV Pilot and Engineer · Collaborative Project	PT Pertamina Hulu Energi West Madura Offshore	Jul 2024 - Aug 2024
- Led technical recovery of a non-operational ROV system: electrical diagnostics, control system restoration, and sea trials - Served as on-site ROV pilot during offshore hull inspection survey - Performed maintenance and troubleshooting on ROV control systems, cameras, and sensors		
Embedded Developer	Penship EEPIS Marine Robotics Team	Mar 2023 - Nov 2024
- Built entire electrical and embedded software systems from zero, progressing from Python+MAVLink to ROS-based autonomy stack - Designed custom PCBs and developed computer vision pipelines (HSV to YOLO) for autonomous navigation on AUV and ASV platforms - Contributed to team achievements in national underwater robotics competitions		
General Manager	Penship EEPIS Marine Robotics Team	Mar 2023 - Dec 2023
- Led 15-member team with 4 robot platforms, establishing weekly meeting culture from scratch - Aligned hardware, software, and mechanical workstreams to competition deadlines		

Hardware Specialist	Penship EEPIS Marine Robotics Team	Mar 2022 - Feb 2023
- Engineered electrical systems for Autonomous Semi Submarine Vehicle: propulsion, power management, and waterproofing - Designed custom PCBs aligned with mechanical specifications for underwater operation - Supported software team with microcontroller programming for mission-based autonomous movements		

Awards and Recognitions

2nd Place , Final Project Competition, Applied Bachelor of Electronics Engineering, PENS	2025
3rd Place National , Underwater Robotics Competition (KRBAI), Kontes Robot Indonesia	2023
2nd Place Region 2 , Underwater Robotics Competition (KRBAI), Kontes Robot Indonesia	2023
National Finalist and Funded Proposal , Pekan Kreativitas Mahasiswa (PKM) Research & Technology, "Jalavahana" ASV Project, Ministry of Education	2024
Most Active and Best Idea , Kaizen Event, PT Aldzama	2026

Certifications

Sea Survival and Offshore Emergency Procedure , PT Global Saftindo (Credential: GS-ADM-024-3594)	Jul 2024 - Jul 2027
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Technical Skills

Embedded Systems: STM32, ESP32, Arduino, AVR, microcontroller peripherals (Digital I/O, ADC, Timer, Counter, Watchdog, UART, SPI, I2C, CAN), real-time control firmware

PCB and Hardware Design: Schematic design and PCB layout (EasyEDA, Eagle, KiCAD), electronics component selection, power electronics, motor drivers, EMC/EMI troubleshooting, soldering and prototyping, oscilloscope and multimeter

Programming: C/C++, Python, JavaScript, Bash

Tools and Workflow: Git (version control), Linux, ROS, OpenCV, YOLO, TensorFlow, MQTT

Other: Electrical wiring and single-line diagram reading, testing and validation, field deployment and troubleshooting

Languages: Bahasa Indonesia (Native), English (Professional Working)